

HEALTH INSURANCE CHARGES ANALYSIS

Impact of Age, BMI, Smoking & Family Size on Medical Costs



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DATE :14 DECEMBER 2025

EXECUTIVE SUMMARY

- Insurance Charges increase with age and BMI.
- Smokers generate significantly higher charges than non-smokers.
- BMI category is a strong predictor of average insurance cost
- Region and sex show minimal impact compared to lifestyle factors

DATASET DESCRIPTION

Individual health insurance records

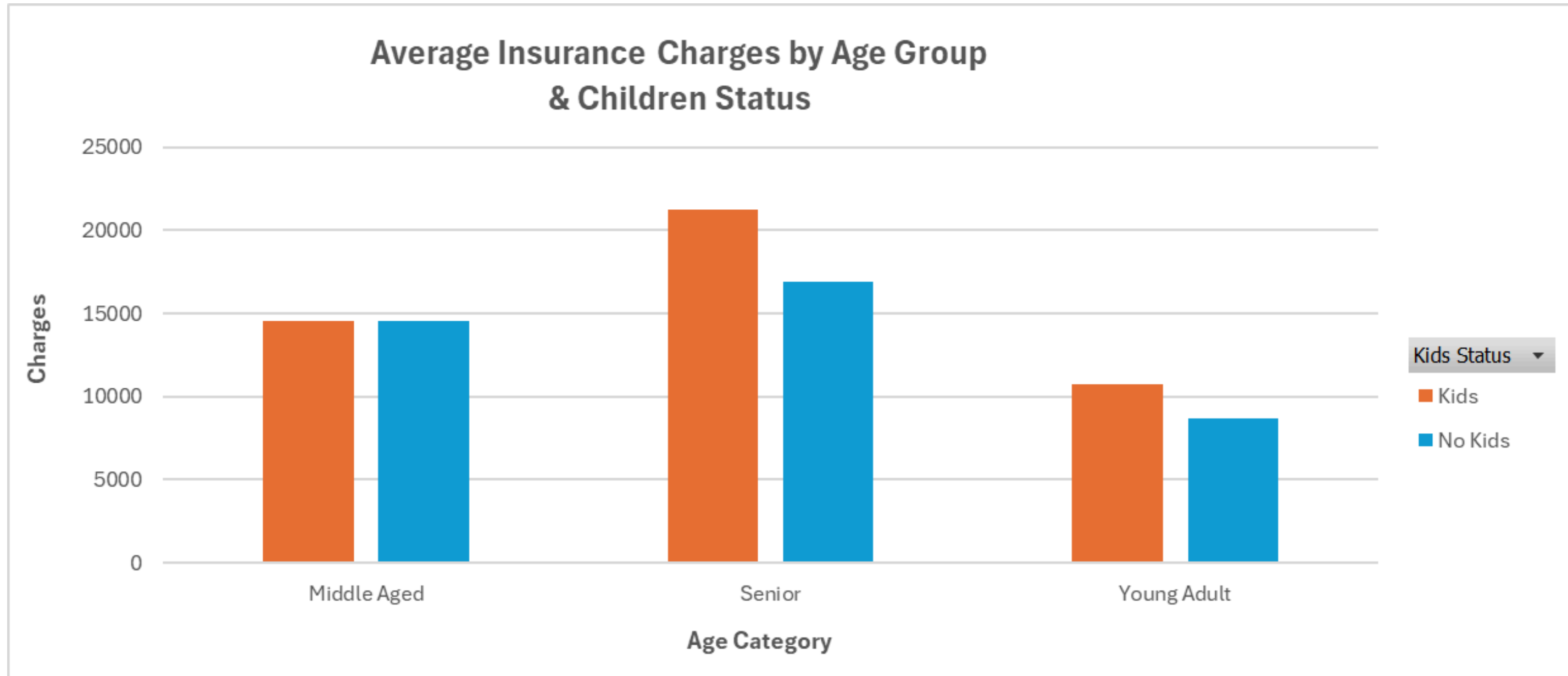
Key Variables:

- Age
- BMI
- Smoking status
- Number of children
- Sex
- Region
- Charges

Focus: Identifying cost drivers in health insurance pricing

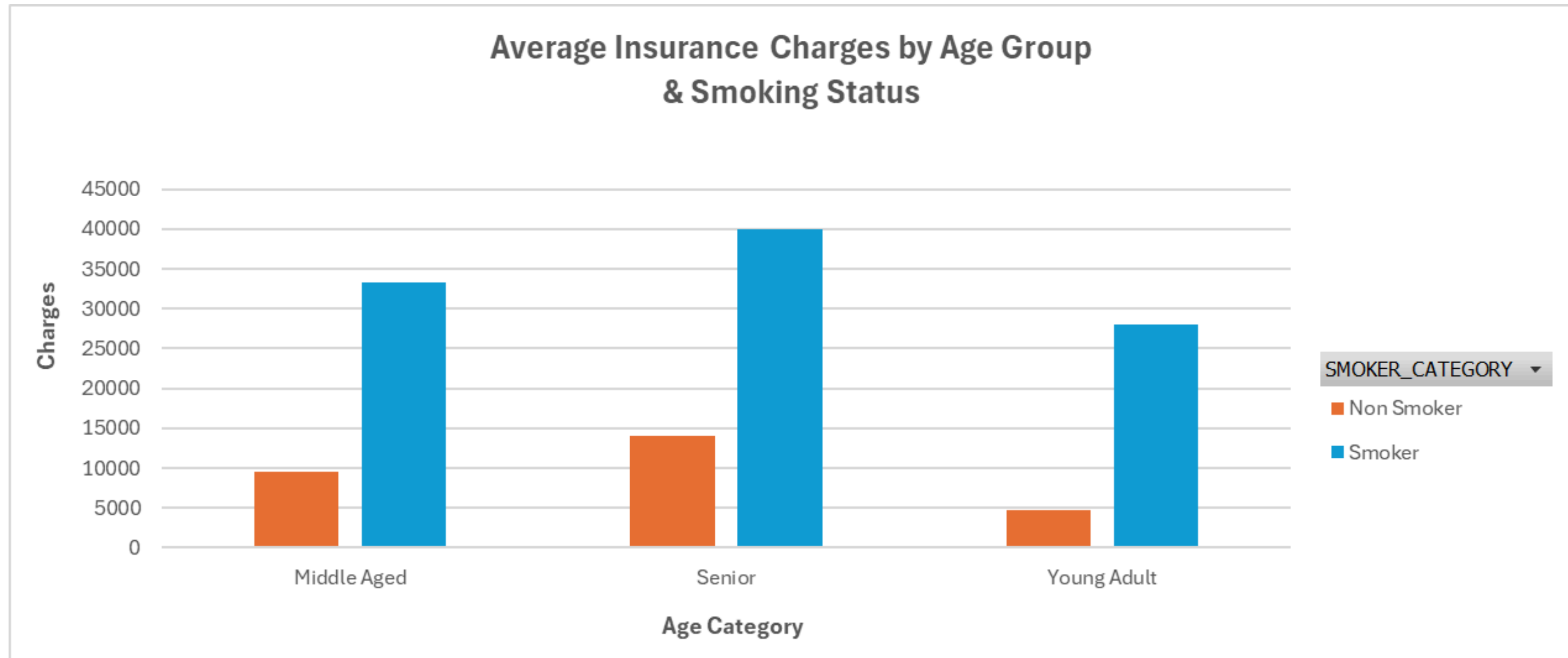
1. AVERAGE CHARGES ANALYSIS

AGE GROUP & CHILDREN STATUS



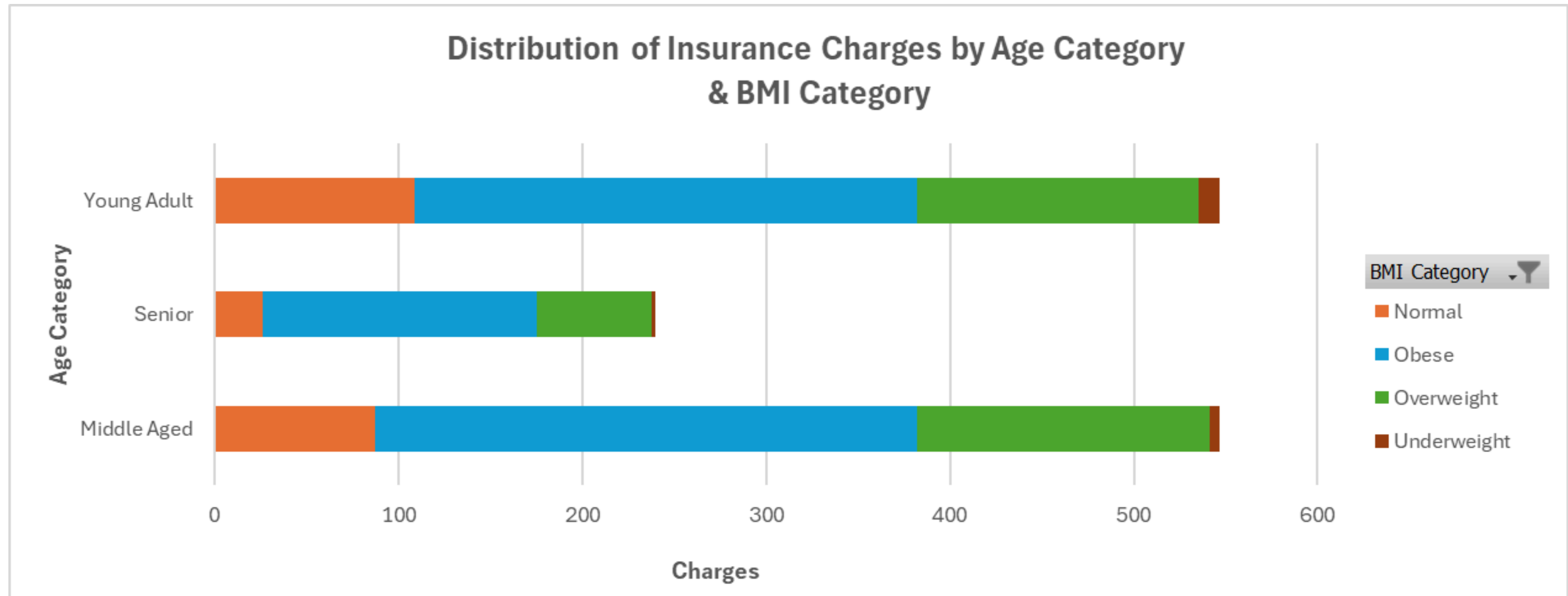
- Seniors pay highest insurance charges (22,000 with kids vs 18,000 without kids.)
- Middle aged people pay 15,000 with kids vs 13,000 without.
- Young Adults pay lowest with 10,000 with kids vs 8,000 for those without.)
- Having kids increases charges across all age groups
- Charges rise significantly with age

AGE GROUP & SMOKING STATUS



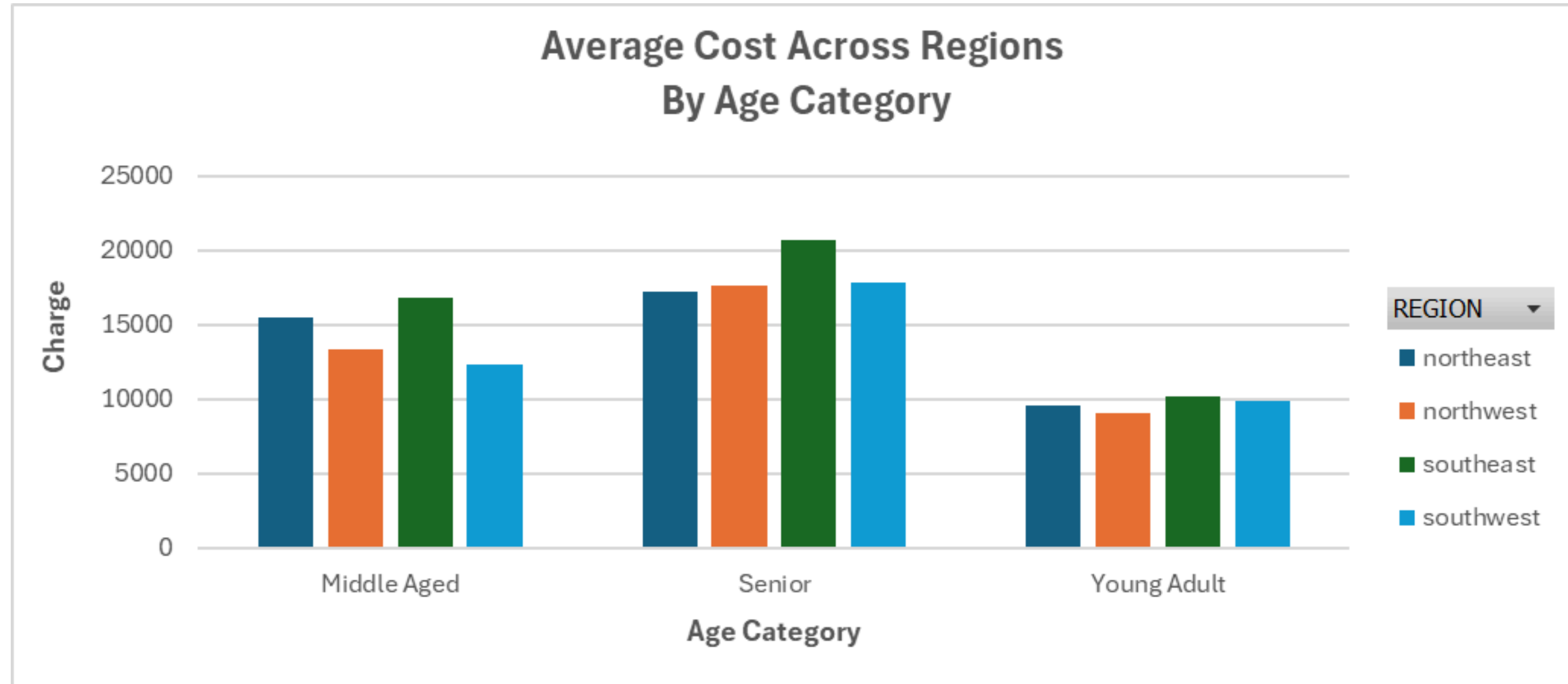
- Smokers pay significantly higher insurance charges than non-smokers across all age groups
- Senior smokers have the highest average charges (around 40,000+).
- Middle-aged and young adult non-smokers have the lowest charges (under 10,000)
- Charges for smokers are consistently 4-10 times higher than for non-smokers in the same age category

AGE & BMI CATEGORY



- Obese (Blue) category has the largest share of insurance charges in all age groups
- Overweight (Green) is the second largest, especially in young Adult and Middle Aged.
- Normal (Orange) and Underweight (dark brown) have the smallest proportions across categories
- Seniors show the most balanced distribution, with smaller obese segment compared to younger groups

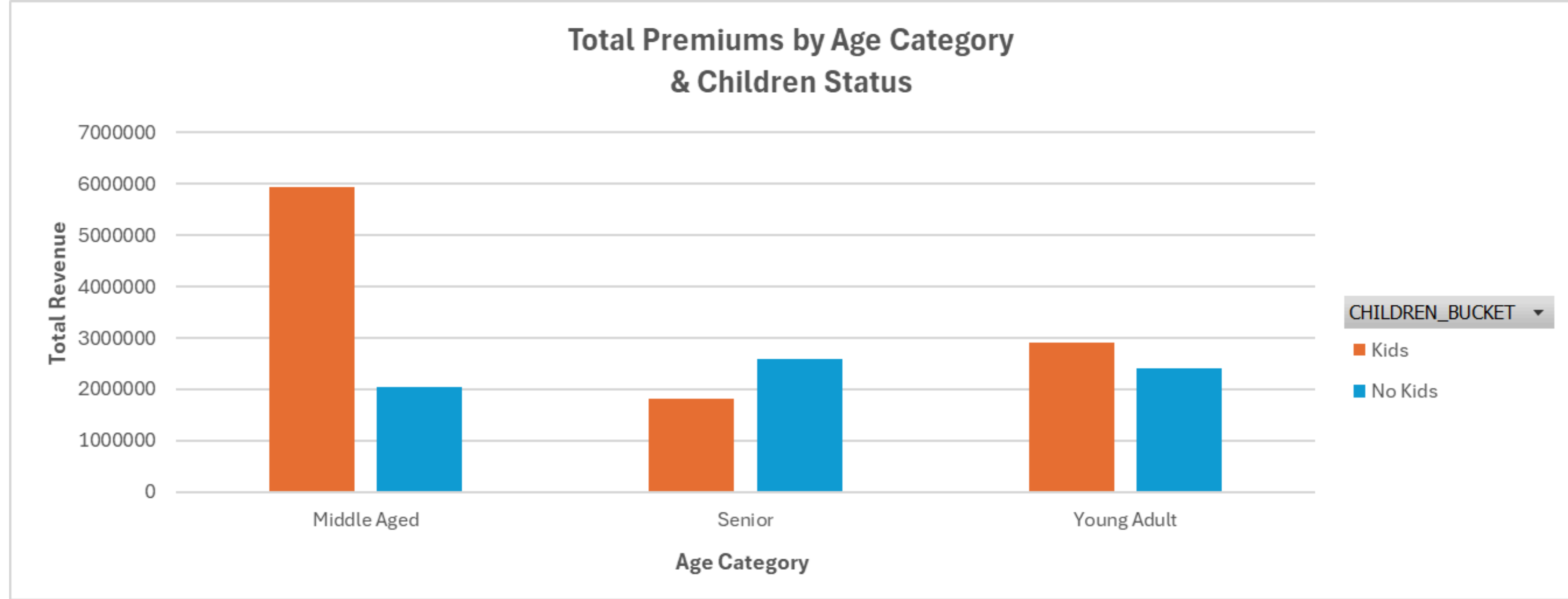
REGIONS & AGE CATEGORY



- Seniors have the highest average insurance costs across all regions (around 15,000-20,000+)
- Young adults have the lowest costs (under 10,000 in all regions).
- Southeast (dark green) often shows the highest costs, especially for seniors
- Costs vary most by region in Seniors; least variation in Young Adults.

2. TOTAL REVENUE ANALYSIS

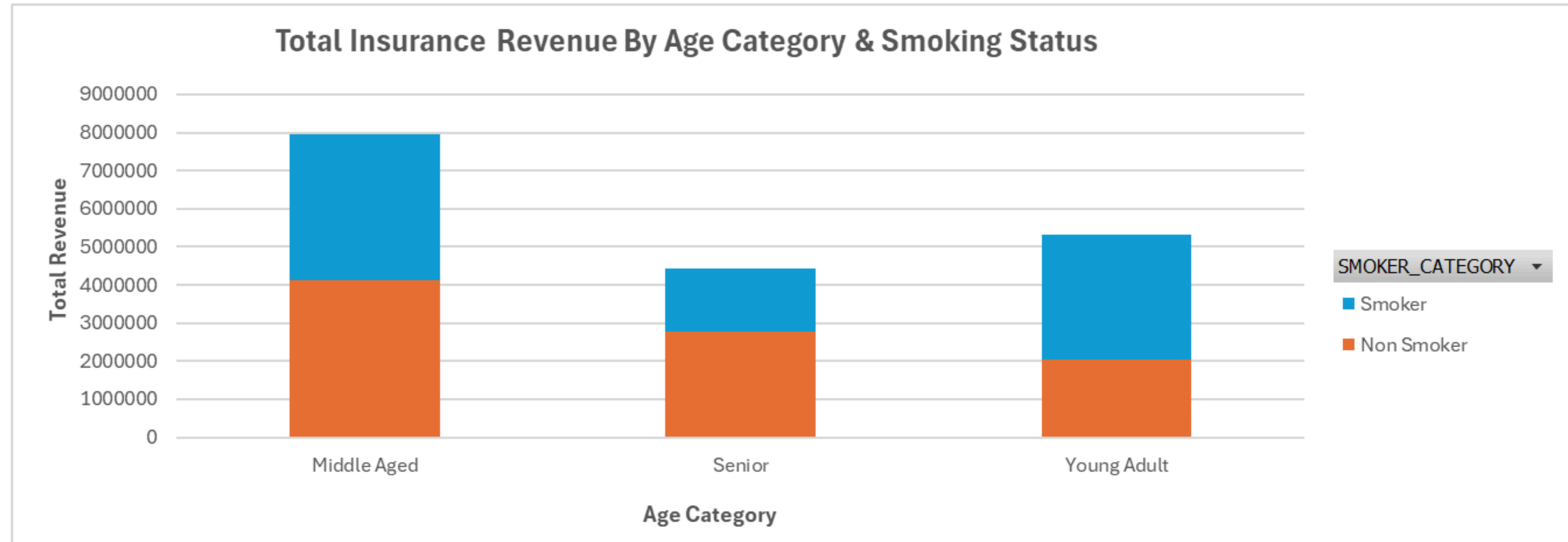
PREMIUMS BY AGE & CATEGORY



- Middle-aged with kids generate highest premiums (600M), far exceeding no-kids(180M)
- Young adults with kids lead their group (250M vs 150M no kids)
- Seniors show lowest premiums, with no kids slightly higher (220M vs 150M with kids)

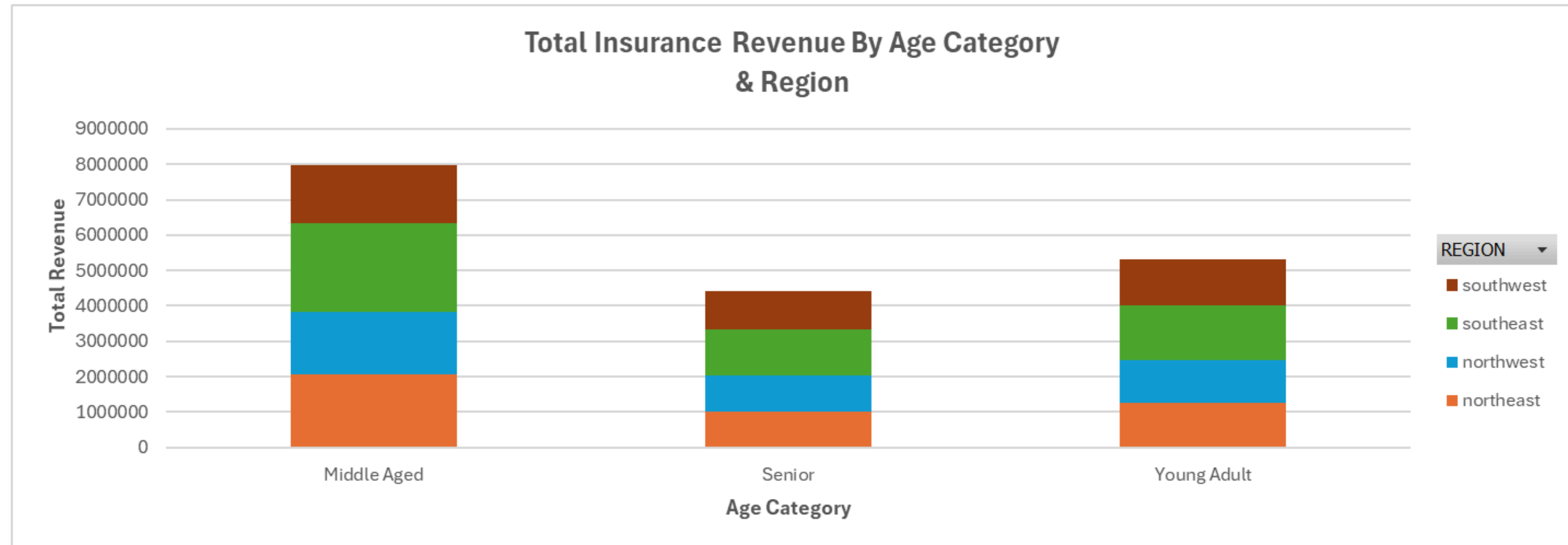
Overall, having children correlates with higher insurance premiums across all age categories.

AGE VS SMOKING STATUS



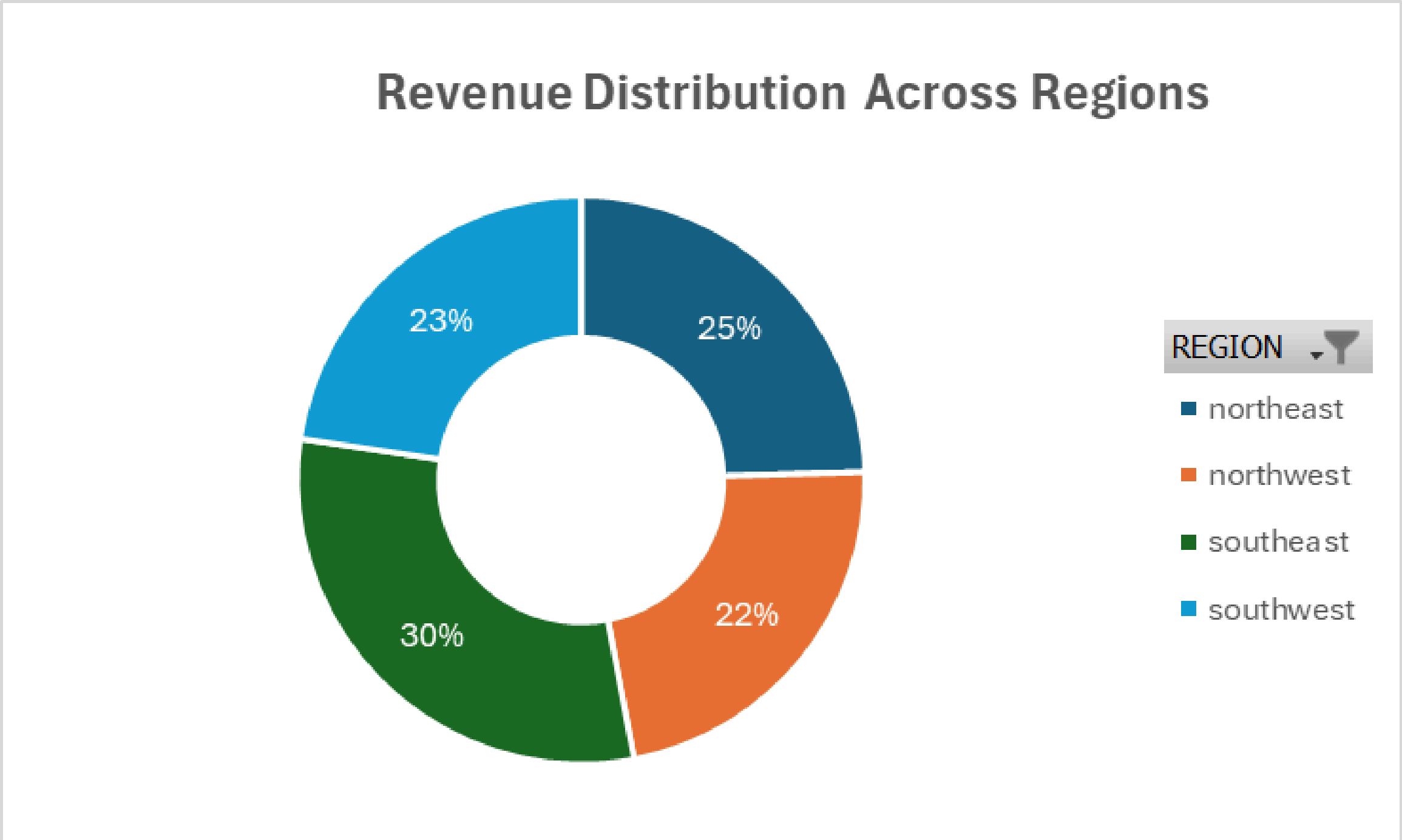
- Middle Aged generates highest revenue (8M), mostly from non-smokers.
- Young Adults follows (5.5M), with smokers contributing more than non-smokers.
- Senior has lowest (4.5M), dominated by non-smokers.
- Non-smokers drive revenue in Middle Aged and Senior.
- Smokers dominate in Young Adults.

AGE, CATEGORY & REGION



- Middle Aged generates highest revenue (8.5M), led by southwest, then southeast, northwest, northeast.
- Young Adults follows (6M), similar regional distribution but southwest dominant.
- Senior has lowest (4.5M), balanced across regions
- Southwest region contributes most across all age groups

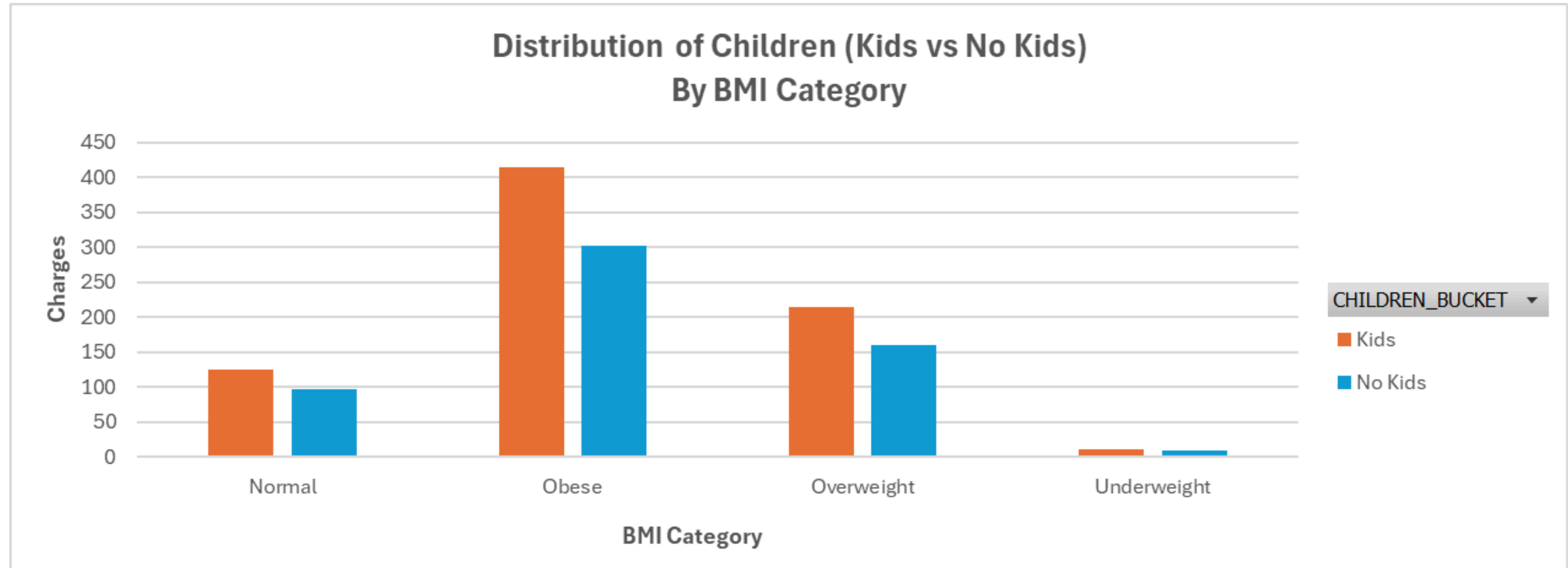
REGIONAL DISTRIBUTION



- Southeast contributes most revenue (30%), followed by Northeast (25%), Southwest (23%), Northwest (22%).
- Regions are closely balanced, with Southeast leading slightly

3. FURTHER CLIENT INSIGHTS

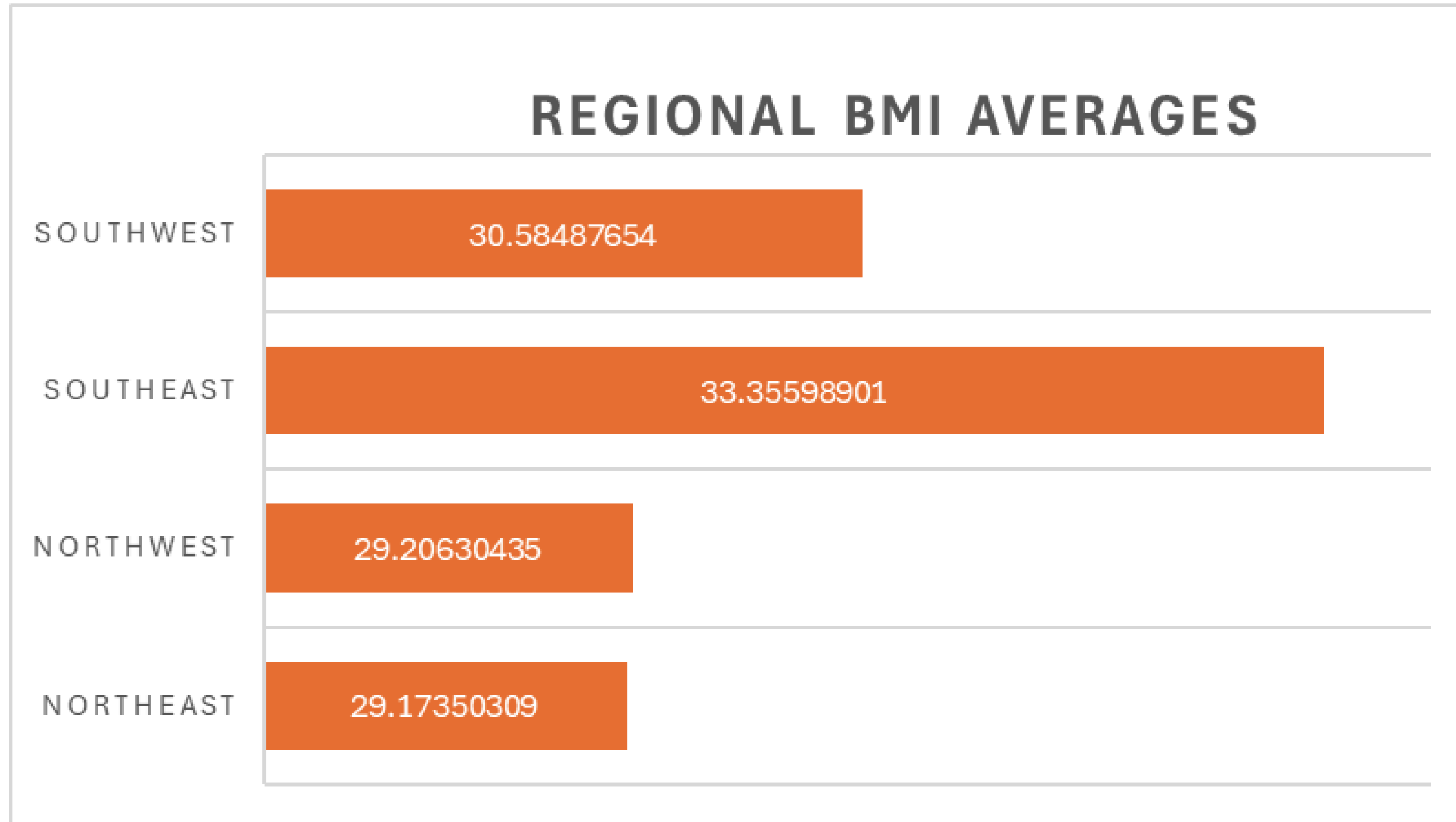
TOTAL (COUNT) DISTRIBUTION



- Obese constitutes the largest group with around 700 total people(400 with kids vs 300 no kids)
- Overweight people have a total of 310, more with kids
- Normal people with equating to 210 slightly more with kids
- Underwight: Negligible

Majority of policy holders are obese/overweight parents

AVERAGE BMI



- Southeast has highest average BMI (33.36), obese range.
- Southwest next (30.58), also obese.
- Northwest(29.21) and Northeast (29.17) overweight
- Southeast significantly higher than others



INSURANCE

THANK YOU

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